Time-domain noise as an evaluation axis for raw-signal nanopore mapping

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Abstract

Motivation: Raw-signal nanopore mappers segment the current trace into events before matching, so their accuracy depends on how event boundaries respond to variation in the time the motor protein holds each k-mer. Mappers are evaluated on real datasets, in which this dwell-time dispersion is fixed by the run, so their sensitivity to it is not measured. Measuring it requires a variable that can be moved, which a signal simulator provides.

Results: We sweep the dwell-time coefficient of variation (CV) from 0 to 0.60 in Squigulator, with a composition-matched control and three seeds. In one quantised-event pipeline, fixed-width segmentation leads event detection at CV 0, 49.7% against 18.4% true-positive rate at zero false positives, and trails it 10.8-fold at the simulator's default CV of 0.44, 1.3% against 14.1%; the replicate ranges do not overlap at any dispersion tested. Evaluation at constant dwell therefore selects the worse configuration. RawHash2 loses 2.9 points at its default setting, so the reversal belongs to the segmentation choice, not to signal-space mapping. RawHash2's minimizer sketching does carry a dwell-dependent cost: paired within seed, the accuracy penalty of window 10 against window 0 rises from 11.7 points at CV 0 to 19.5 at CV 0.60, and is larger at CV 0.60 in every seed.

Availability and implementation: Code, result files, a patch adding dwell variation to the Icarust simulator, and a script that recomputes every number in this article: `https://github.com/khushal6001-maker/microsecond-readuntil`, archived at `https://doi.org/10.5281/zenodo.23604395` (release v1.0.0).

Supplementary information: Supplementary data are available at *Bioinformatics* online.

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1 Introduction

Signal-space mappers align a nanopore current trace to a reference without basecalling. UNCALLED (Kovaka et al., 2021) matches events against an FM-index of the reference; Sigmap (Zhang et al., 2021) indexes event-value seeds in a k-d tree and chains them; RawHash (Firtina et al., 2023) and RawHash2 (Firtina et al., 2024) hash quantised event k-mers and chain the resulting anchors; and Sigmoni (Shivakumar et al., 2024) classifies reads with an r-index over a small current alphabet. Their principal use is where a decision must be made while the molecule is still in the pore, as in adaptive sampling (Kovaka et al., 2021; Payne et al., 2021).

Every pipeline in this family begins by dividing the trace into events, each intended to correspond to one k-mer. The order of stages matters. A key is built from consecutive events, so a query boundary that does not correspond to one implied by the reference changes the key, and no later stage can restore it. Event duration is set by the motor protein, whose stepping is stochastic: the number of samples per k-mer varies from one k-mer to the next, and its distribution is right-skewed. On real signal the variation is large enough that signal aligners model it explicitly: Uncalled4 attributes frequent stays, consecutive events that represent the same k-mer, to variable sequencing speed (Kovaka et al., 2025), and SegPore builds the motor protein's dynamics into segmentation itself (Cheng et al., 2025). The accuracy of a segmentation-based mapper is therefore a function of dwell-time dispersion. What is not known, for any published pipeline or parameter choice, is how strong that dependence is.

Evaluation practice does not measure it. Mappers are evaluated chiefly on real datasets, and RawBench (Eris et al., 2025), which decomposes raw-signal analysis into reference encoding, signal encoding and representation matching and evaluates combinations of them, does so on real datasets as well. In a real dataset dispersion is a property of the run. The experimenter neither sets it nor, in general, reports it, and two datasets differ in more than dispersion. Each published accuracy figure is therefore one point at an unknown position on an axis whose slope has not been measured. CRANE (Ambrozak et al., 2026) corrects errors in raw signal before analysis and reports gains for existing raw-signal tools, which shows directly that signal-level error propagates to mapping accuracy; it does not isolate time-domain noise as a variable.

Why simulation is the instrument. A sensitivity is a derivative, and measuring one requires moving the variable while everything else is held. A fixed biological dataset supplies a single value of dwell dispersion, which cannot be changed without also changing the sample, the pore, the chemistry or the run. Squigulator (Gamaarachchi et al., 2024) exposes dispersion directly, as the standard deviation of the number of samples per base, with a documented default of mean 9.0 and standard deviation 4.0 samples, a CV of approximately 0.44. For this question simulation is not a substitute for real data but the only instrument with the required control. What simulation cannot establish, the correspondence between a simulator setting and the dispersion of a particular device or run, is stated as a limitation in Section 5 rather than assumed.

The question has a practical edge. Icarust (Munro et al., 2024), the simulator that serves the MinKNOW read-until interface and so allows a client to be tested end to end, emits a fixed number of samples for every k-mer, which is the CV = 0 case exactly. The one setting in which a read-until client can be exercised end to end is therefore the one setting in which a dwell-sensitive failure cannot appear.

This article makes five contributions.

1. A protocol for sweeping dwell dispersion as an evaluation variable, with a composition-matched negative control simulated at matching dwell, three generator seeds, and three compatibility conditions, each of which, when violated, produces a plausible curve that measures something else (Section 2).

2. A ranking reversal: two segmentation configurations of one pipeline exchange order between CV 0 and CV 0.10, so that an evaluation at constant dwell selects the configuration that is 10.8-fold worse at the simulator's default (Section 3.1).

3. A bounding result: RawHash2 does not show the reversal, which locates it in the segmentation choice rather than in the method class (Section 3.2).

4. A dwell-dependent accuracy cost of minimizer sketching in RawHash2, which we have not found measured elsewhere (Section 3.3).

5. A patch that adds lognormal dwell variation to Icarust, so that live read-until testing can exercise the axis (Section 3.4).

What this article does not claim. It does not claim that published mappers segment by fixed time windows; none of those surveyed does. The fixed-width arm is a configuration of the authors' own pipeline, included because it is the configuration a constant-dwell evaluation favours. It does not claim that signal simulators omit dwell variation: Squigulator models it, and the gap identified here is specific to Icarust. It does not claim that the authors' pipeline is competitive. At CV 0.44 its better configuration resolves 14.1% of reads at zero false positives, where RawHash2 maps 91.8% correctly at its default operating point, and every comparison drawn below concerns the shape of a system's response to dispersion, not its level.

2 Methods

2.1 Signal generation

Reads were simulated with Squigulator under its R10.4 PromethION DNA profile (`dna-r10-prom`), which samples at 5 kHz. The dwell mean was fixed at 10.0 samples per base and the dwell standard deviation swept over $\{0, 1, 2, 3, 4.4, 6\}$ samples, giving CV $\in \{0.00, 0.10, 0.20, 0.30, 0.44, 0.60\}$. The mean is held fixed so that only dispersion varies; a varying mean would also change the number of bases a chunk of fixed duration contains, and so confound dispersion with the quantity of evidence at the decision point. Each cell was simulated with three generator seeds (42, 43 and 44) and contains 500 on-target reads and 500 control reads.

2.2 Reference, pore model and control

The reference is a 20 Mb prefix of human chromosome 20 (GRCh38), indexed together with its reverse complement, 40 Mb of encoded sequence in all. The pore model is Squigulator's own R10.4 9-mer table, extracted from its source, and is used both to generate signal and to encode the reference in both systems under test. The negative control is a composition-matched shuffle of the same reference, simulated at the same dwell setting as the on-target set in each cell. Composition and dwell are therefore identical between the two sets, and the only difference between them is whether the sequence is present in the index.

2.3 Three compatibility conditions

Each of the following was violated in a first attempt during this work, and each violation produced a smooth, plausible curve. They are stated because they apply to any comparison of this kind.

The pore model must be the same in the generator and in both encoders. The R10 9-mer table shipped with Icarust and the table built into Squigulator differ, so encoding the reference with one while generating signal from the other measures model mismatch rather than dispersion.

Both strands must be indexed. Squigulator samples reads from both strands, while reference encoding is defined on the forward sequence. Indexing one strand makes about half of all reads unmappable by construction and halves every true-positive rate for a reason unrelated to the axis.

Chunk length must be defined in time. The decision procedure consumes chunks of fixed duration. At this profile's 5 kHz, 0.4 s is 2000 samples; a sample count carried over from a profile with a different sampling rate misaligns the decision point and changes the evidence available at it.

A sensitivity curve is meaningful only to the extent that everything other than the swept variable is invariant, and these are invariances that no error message reports.

2.4 Configurations under test

Both arms are configurations of one quantised-event read-until pipeline written by the authors, and both are available in the repository. The *fixed-width* arm divides the trace into windows of a constant number of samples and builds keys of 14 events at 3 bits per event, sketched with minimizer window 10. The *detected* arm segments with a two-window t-test changepoint detector (window 4 samples, threshold 3.0, minimum event length 4 samples) and builds keys of 12 events at 4 bits per event, with minimizer window 1.

Both decide after 10 chunks of 2000 samples, 0.4 s each, with signal scaling estimated from the first chunk, as the pipeline does in live operation. The detector threshold was calibrated on Squigulator signal (Supplementary Section S2).

The reported statistic is the true-positive rate (TPR) at the lowest score threshold that admits no control read, that is, at zero false positives. This threshold is set by the single highest-scoring control read, so it has high variance when the TPR is low; for that reason we report the range over seeds alongside the mean.

2.5 Comparison mapper

RawHash2 v2.1 was run on the same SLOW5 files, given the same pore model through `-p`, `-k 9` and `--level_column 1`, and otherwise run at its defaults (`-q 4 -e 8 --sig-diff 0.35`) except for the minimizer window, $w \in \{0, 5, 10\}$; $w = 0$, sketching off, is the default. A read is counted as correctly mapped if it maps within 500 bases of the locus Squigulator records in the read identifier, on the matching strand, and the false-mapping rate is the percentage of control reads that receive any mapping.

Two differences from Section 2.4 limit what may be compared. RawHash2's default operating point admits false mappings, so its correct-mapping rate is not level-comparable with a zero-false-positive TPR; the comparison is of shape. RawHash2 also receives whole reads rather than the first 10 chunks, which favours it. The dwell × window grid was run for the same three seeds as Table 1, with 500 reads per set, by `scripts/rerun_rawhash_grid.sh`; seed 42 reproduces an earlier single-seed run of the grid exactly, in every cell. At its default and at CV 0.44, RawHash2 maps 91.8% [91.2–92.4] of reads correctly and 9.1% [8.4–10.6] of control reads.

2.6 Per-event agreement

To test whether a flat accuracy curve reflects a flat encoding, we measured agreement between reference and query symbol sequences for both systems with one estimator. Each system's quantised symbols were dumped for the read and for the reference span at its true locus. For RawHash2 the read side comes from its own `--out-quantize` option; the reference side comes from `--out-quantize-ref`, which a patch supplied in the repository adds, since the stock reference path emits no symbols. The two sequences were aligned, agreement was taken as matches over aligned pairs under the maximising alignment, and the indel fraction as gap columns over all alignment columns. Both of this pipeline's segmentations were measured with keys of 12 events at 4 bits per event, so the fixed-width row isolates the segmentation change from the 3-bit key of the fixed-width arm in Table 1, and scaling was estimated from the whole read. The measurement covers 200 reads per cell, at CV 0 and CV 0.44 only, with one seed.

2.7 Reproducibility

Every number in this article is recomputed from the committed result files by `scripts/verify_dwell_journal_numbers.py`, which uses decimal arithmetic with round-half-up, checks every numeral in this text, in its Markdown source and in the Supplementary Information against the recomputed set, and exits non-zero on any disagreement. Three observations made during development, on live accept rates against Icarust, on the calibration target of the event detector and on the transfer of a detector calibration between signal generators, rest in part on runs whose output was not captured; they are reported, marked, in the Supplementary Information and are used in no claim here. Figures are drawn by `scripts/generate_thesis_figures.py` from the same files and from nothing else.

3 Results

3.1 Constant dwell reverses the ranking of two segmentation configurations

Table 1 and Figure 1 give the TPR of both arms across the sweep. The fixed-width arm falls from 49.7% at CV 0 to 1.3% at CV 0.44, a 38.2-fold reduction, and is already reduced 8.4-fold at CV 0.10, the smallest

Table 1. True-positive rate (%) at zero false positives, mean [range] over three seeds, 500 on-target and 500 control reads per cell.

Dwell CV	Fixed-width (3×14, $w = 10$)	Detected (4×12, $w = 1$)
0.00	49.7 [49.2–50.2]	18.4 [17.0–19.8]
0.10	5.9 [5.4–6.6]	19.3 [15.2–23.0]
0.20	1.7 [1.2–2.0]	17.9 [17.2–19.0]
0.30	2.3 [1.4–4.2]	15.5 [12.6–18.8]
0.44	1.3 [1.0–1.6]	14.1 [13.0–15.0]
0.60	1.7 [0.4–3.4]	7.5 [7.2–8.0]

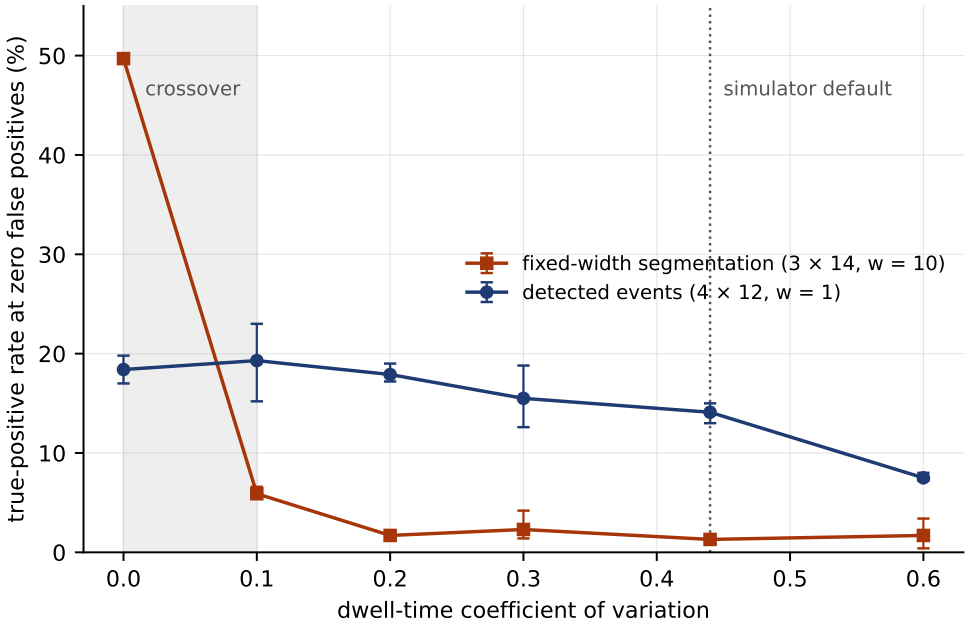


Figure 1. True-positive rate at zero false positives against dwell-time CV for the two segmentation configurations. Bars span the observed range over three seeds. The shaded band marks the interval in which the two curves cross; the dotted line marks Squigulator's default dispersion.

non-zero dispersion tested. The detected arm remains between 14.1% and 19.3% up to CV 0.44 and declines to 7.5% at CV 0.60.

The two arms exchange order between CV 0 and CV 0.10. The replicate ranges do not overlap at any dispersion tested: at CV 0 the lowest fixed-width replicate, 49.2, exceeds the highest detected replicate, 19.8, and at every non-zero dispersion the highest fixed-width replicate falls below the lowest detected one. The order therefore holds in every seed and not only in the means.

The consequence is for configuration selection. An evaluation conducted at CV 0 selects the fixed-width arm, which leads by 2.7-fold there and trails by 10.8-fold at CV 0.44. This is not an error of the kind that more data would correct, because every measurement available at CV 0 supports it. The fixed-width arm is not a published method, and the finding concerns the choice between two configurations of one pipeline under a mis-specified evaluation. Section 3.2 shows that it does not extend to the method class.

3.2 A published mapper does not reverse

At its default, $w = 0$, RawHash2 loses 2.9 points [2.8–3.0] of correct mapping between CV 0 and CV 0.60, with no crossover and no collapse (Table 2, Figure 2). The sensitivity found in Section 3.1 is therefore a property of fixed-width segmentation and not of quantised-event matching, and a developer evaluating RawHash2 at constant dwell would reach the same conclusion as one evaluating it at realistic dispersion.

Table 2. RawHash2 across the dwell sweep: percentage of on-target reads correctly mapped and percentage of control reads mapped at all, by minimizer window w. Mean over three seeds, 500 reads per set; Figure 2 shows the range across seeds, and `results/rawhash-grid-seeds.csv` every seed.

		Dwell CV					
Rate	w	0.00	0.10	0.20	0.30	0.44	0.60
Correct	0	93.5	93.8	93.3	92.7	91.8	90.5
Correct	5	88.8	88.4	88.7	87.4	85.4	81.8
Correct	10	81.8	82.1	82.1	80.5	76.4	71.1
False-mapped	0	10.1	11.1	8.1	10.4	9.1	10.5
False-mapped	5	9.8	10.7	8.5	9.5	9.9	10.7
False-mapped	10	14.4	12.9	13.9	13.7	14.3	15.8

Table 3. Per-event agreement and indel fraction between reference and query symbol sequences, one estimator applied to both systems, 200 reads per cell, one seed.

System	Segmentation	Dwell CV	Agreement	Indel fraction
This pipeline	fixed-width	0.00	0.793	0.073
This pipeline	fixed-width	0.44	0.648	0.208
This pipeline	detected	0.00	0.797	0.170
This pipeline	detected	0.44	0.783	0.196
RawHash2	own, with merging	0.00	0.913	0.055
RawHash2	own, with merging	0.44	0.911	0.092

With sketching enabled the decline is steeper, 7.0 points [5.2–8.4] at $w = 5$ and 10.7 [9.4–11.6] at $w = 10$, which is the subject of Section 3.3.

The response to dispersion is visible in the encoding before any seeding or chaining (Table 3). RawHash2's per-event agreement is 0.913 at CV 0 and 0.911 at CV 0.44, and that of the detected arm 0.797 and 0.783. The fixed-width segmentation's agreement falls from 0.793 to 0.648, and its indel fraction rises 2.8-fold, from 0.073 to 0.208. RawHash2's indel fraction also rises, from 0.055 to 0.092, without a corresponding loss of agreement. Both segmenters that detect level changes thus have dispersion-invariant agreement, and the one that segments by time does not. These are two dwell levels, 200 reads and one seed (Section 5).

3.3 The cost of minimizer sketching depends on dwell dispersion

Table 4 and Figure 3 give the accuracy cost of sketching: the correct-mapping rate at $w = 0$ minus that at a larger window, in percentage points. Every window of one seed maps the same reads, so the cost is taken within each seed and then summarised over the three. At $w = 10$ it is 11.7 points [10.4–12.8] at CV 0 and 19.5 [18.2–21.0] at CV 0.60. The two ranges do not overlap, and the cost is larger at CV 0.60 than at CV 0 in every seed, by 7.8 points on average [6.4–8.8]. At $w = 5$ it goes from 4.7 [4.0–5.0] to 8.7 [7.2–10.4], again larger in every seed. Sketching at $w = 10$ also maps more control reads than $w = 0$ at every dispersion and in every seed (Table 2), so the sensitivity it gives up is not returned as specificity.

The shape, and the limits of the claim. Up to CV 0.30 the mean $w = 10$ cost stays between 11.2 and 12.3 points, and each of those cells' ranges overlaps the CV 0 range. At CV 0.44, the simulator's default, it is 15.4 [14.6–16.4], a range already clear of the CV 0 range, and it rises again at CV 0.60. The claim made is that the cost is larger at realistic and high dispersion than at constant dwell, in every seed; it is not that the cost rises at every step. Three seeds give an observed range rather than a confidence interval, which is why each claim in this section is one that holds in every seed separately.

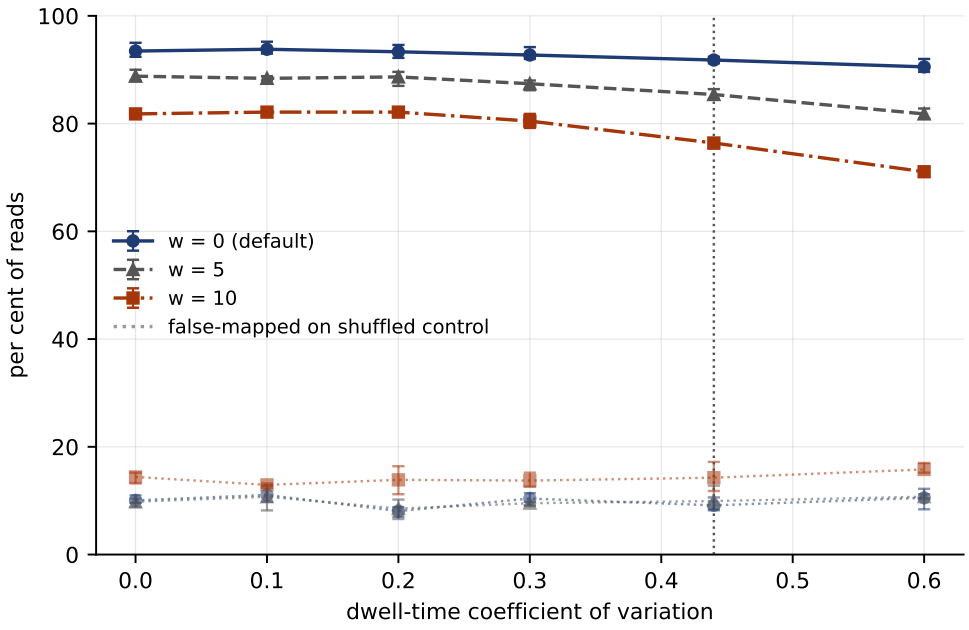


Figure 2. RawHash2 under the dwell sweep. Solid, dashed and dash-dotted lines: percentage of on-target reads correctly mapped, by minimizer window. Dotted lines: percentage of control reads mapped. Points are means over three seeds and bars span the observed range. The default operating point is not zero-false-positive, so these rates are not level-comparable with Figure 1.

Table 4. Accuracy cost of minimizer sketching in RawHash2: correct-mapping rate at $w = 0$ minus that at the stated window, in percentage points, taken within each seed; mean [range] over three seeds.

Dwell CV	$w = 5$	$w = 10$
0.00	4.7 [4.0–5.0]	11.7 [10.4–12.8]
0.10	5.4 [4.6–6.4]	11.7 [10.6–13.0]
0.20	4.7 [3.6–5.2]	11.2 [10.4–12.4]
0.30	5.3 [4.0–6.2]	12.3 [10.2–15.0]
0.44	6.4 [5.4–7.2]	15.4 [14.6–16.4]
0.60	8.7 [7.2–10.4]	19.5 [18.2–21.0]

What is known and what is added. RawHash2 ships with sketching disabled to obtain its highest accuracy, and reports that its minimizer mode maps with accuracy similar to the default except on the human genome, at higher throughput (Firtina et al., 2024). That a trade-off exists is therefore known and documented. What the sweep adds is that its accuracy side is a function of dispersion. A window chosen from a comparison at one dispersion carries a different cost at another, and a comparison on constant-dwell signal understates it.

A candidate mechanism, not isolated. A minimizer is the key of lowest rank within a window of consecutive keys (Roberts et al., 2004). A displaced event boundary changes the quantised values of the keys that overlap it and can therefore change which key in a window attains the minimum, so that reference and query select different keys even where both would match individually. This requires boundaries to move. At constant dwell the cost reflects only the seeds that sketching discards; the increment above it is consistent with the mechanism, which would be demonstrated by recording which keys each side selects, and that was not done.

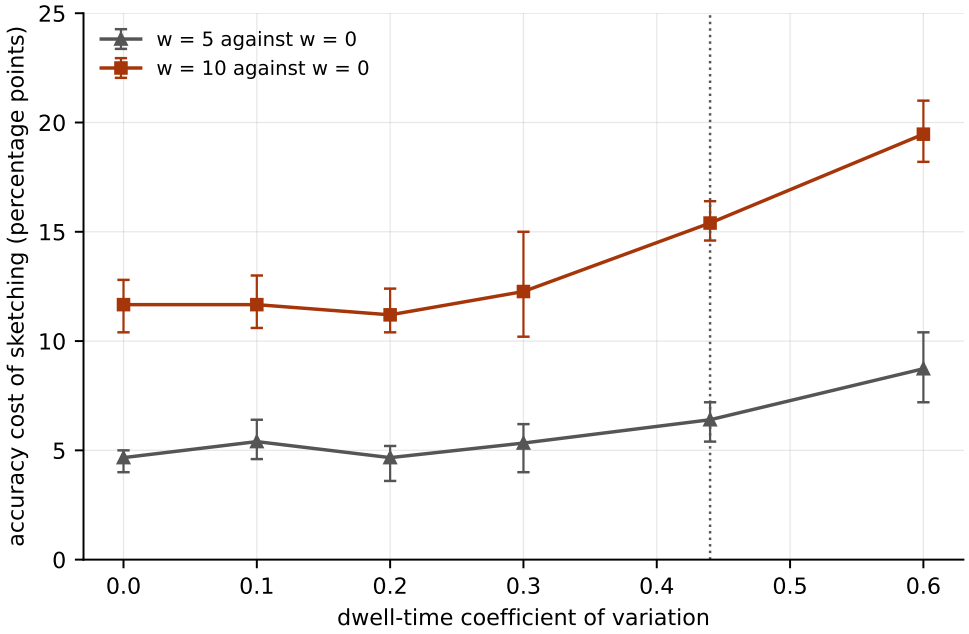


Figure 3. Accuracy cost of minimizer sketching in RawHash2 against dwell-time CV, for windows 5 and 10 relative to window 0, taken within each seed. Points are means over three seeds and bars span the observed range. The dotted line marks Squigulator's default dispersion.

3.4 Icarust emits constant dwell, and a patch that removes the restriction

Icarust's signal loop emits `samples_per_base` samples for every k-mer, and a source comment notes that the count could instead be derived from the profile's dwell mean. Its output is therefore the CV = 0 case exactly, and no configuration of the stock simulator produces time-domain noise.

The patch in `third_party_patches/icarust-dwell-variability.patch` reads a CV from the environment variable `ICARUST_DWELL_CV` and draws each k-mer's dwell from a lognormal distribution about `samples_per_base`, with $\sigma^2 = \ln(1 + \mathrm{CV}^2)$ and $\mu = \ln(\texttt{samples_per_base}) - \sigma^2/2$, so that the mean before rounding equals `samples_per_base` and the CV equals the requested value. Each draw is rounded and clamped to at least 1 sample, since a k-mer contributing nothing is a skip, which is a different phenomenon, and to at most 8 times nominal, beyond which a hold is a pore block rather than a dwell. Unset or 0, the variable reproduces the stock behaviour exactly, so existing results are unaffected. An environment variable keeps the change minimal and revertible; a version offered upstream should read the value from the sample profile.

4 Discussion

The sweep makes visible a variable that a point evaluation holds fixed at an unknown value, and in the two cases measured here the variable decides the outcome. In Section 3.1 it decides which of two configurations is better; in Section 3.3 it decides how much a documented trade-off costs. Neither result depends on the authors' pipeline being accurate, since the first concerns the relative order of two configurations and the second is measured on RawHash2.

We draw six recommendations for evaluating raw-signal methods.

1. **Report dispersion with accuracy.** For simulated signal, report the generator's dwell parameters. For real signal, report an estimate of dwell dispersion, which could be obtained from basecaller move tables, since these record the samples assigned to each base.

2. **Report a sweep, not a point,** for any method or parameter that depends on segmentation.

3. **Compare configurations at the dispersion of the intended data.** A comparison at CV 0 can reverse a ranking (Section 3.1).

4. **Treat the sketching window as dispersion-dependent,** and measure its cost at the dispersion of the intended data (Section 3.3).

5. **State the calibration target and calibration signal of any event detector,** and calibrate against agreement rather than event count (Supplementary Section S2).

6. **Test read-until clients against a simulator with a dwell model** (Section 3.4).

The sweep is complementary to existing benchmarks. RawBench varies the components of the pipeline on fixed data; the sweep varies one property of the data with the pipeline fixed, and adding a simulated-dispersion axis to a modular benchmark would combine the two. It is equally a way to quantify methods that correct or model the signal before segmentation: the benefit of CRANE's correction, or of SegPore's model of motor-protein dynamics, could be reported as a function of the noise source it addresses rather than at one point. Any pipeline that builds keys from consecutive events, whether by hashing quantised values or by seeding on event vectors, is exposed to boundary placement in the way Section 3.3 describes, and the sweep is an inexpensive way to measure that exposure.

5 Limitations

One reference and one profile. All figures derive from a 20 Mb prefix of one human chromosome under one R10.4 profile. Reference size affects specificity and hence absolute rates; the claim is the shape of the response to dispersion, and it has not been tested on a second reference.

Simulated signal only. The mapping from Squigulator's dwell parameter to the dispersion of any particular device or run is not established here. Measuring the empirical dispersion of a public dataset and marking it on the axis is the most direct extension.

Replication. Every accuracy result is over three generator seeds of 500 reads per set. Three seeds give an observed range, not a confidence interval, so the claims made are those that hold in each seed separately; they bound variation between simulated read sets, not between sequencing runs.

Mechanisms. The minimizer mechanism of Section 3.3 is consistent with the data and not isolated. The agreement measurement of Section 2.6 covers two dwell levels, 200 reads and one seed.

Absolute level of the detected arm. The detected configuration's indel fraction is 0.170 already at CV 0, where there is no timing noise to explain it, which indicates a structural mismatch between this pipeline's reference and query encodings. It lowers that arm's absolute TPR, and it is one reason we draw only comparisons of shape.

Operating points. The two systems are measured at different operating points, zero false positives for the authors' pipeline and RawHash2's default for RawHash2.

Scope of the reversal. The fixed-width arm is a configuration of the authors' pipeline. No published tool's evaluation is claimed to be wrong.

6 Conclusion

Dwell-time dispersion is a signal property that Squigulator exposes as a parameter and that published evaluations of raw-signal mappers do not vary. Sweeping it, with a composition-matched control and three replicates, shows that an evaluation at constant dwell can select a segmentation configuration that is 10.8-fold worse at realistic dispersion, and that the accuracy cost of minimizer sketching in RawHash2 rises from 11.7 to 19.5 points over the range swept, in every seed. The first result is bounded by RawHash2's flat response and the second is measured on it. Both follow from holding fixed a variable that evaluation can move, and both are avoided by reporting accuracy as a function of it.

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Conflict of interest

None declared.

Data availability

All result files, raw logs, the sweep harness, the figure generator, the Icarust and RawHash2 patches and the verification script are in the repository given under Availability and implementation, archived at `https://doi.org/10.5281/zenodo.23604395` (release v1.0.0).

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